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Unit - 5
Assignment - 5

Q1 Why do substitution reaction occur in benzene?

Ans Substitution reaction occur in benzene because they preserve its aromatic stability. Addition reactions would break the ~~decol~~ delocalized π-e^- system & make the molecule less stable so benzene prefers electrophilic substitution instead.

Q2 What are electrophile & nucleophiles give examples also?

Ans • Electrophiles are species that love electrons. They are electron-deficient & seek electrons.

→ Think: Electron loving (accept electrons)

→ Example - H^+, NO_2^+, Br_2, $AlCl_3$

• Nucleophiles are species that loves nuclei (positive charge). They are electron-rich & donate electrons.

→ Think - Nucleus loving (donate electrons)

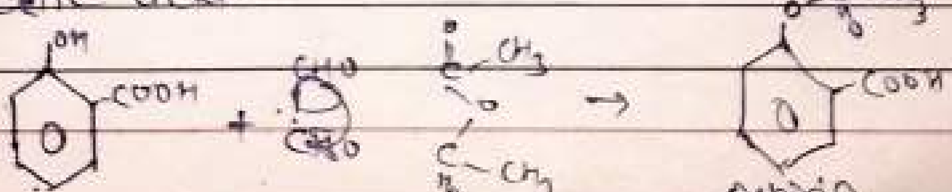
→ Example - OH^-, NH_3, Cl^-, H_2O.

Q3 Describe synthesis & properties of Aspirin & ~~protoc~~ paracetamol.

Ans

Aspirin :-

Synthesis - Salicylic Acid reacts with acetic anhydride in the presence of sulfuric acid to form aspirin & acetic acid.



Aspirin

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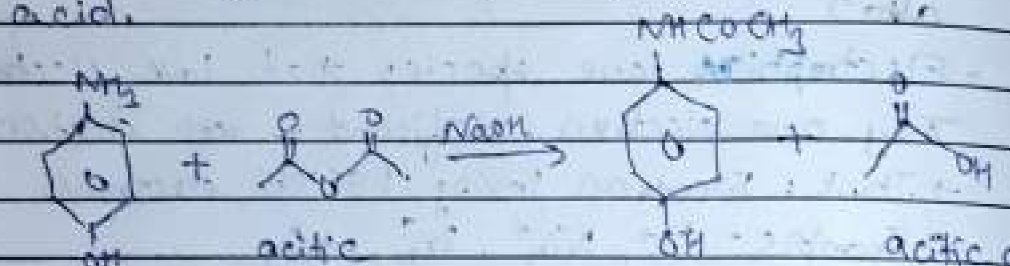
Properties - White crystalline solid
- Melting point - 136-140°C
- Slightly soluble in water
- Used as painkiller, anti-inflammatory & blood thinner.

Paracetamol :- (Acetaminophen)

Synthesis - p-aminophenol reacts with acetic anhydride to form paracetamol & acetic acid.



Para amino phenol. | acetic Anhydride | Acetaminophen | acetic acid

Properities -
- white crystalline powder
- Melting point 169° - 172°C
- Soluble in hot water
- Used as pain & fever reliever (milder than aspirin)

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Q1 Explain mechanism of electrophilic & free radical addition in alkene.

Ans Electrophilic Addition:-

(i) Electrophilic attacks the electron rich double bond (π bond) of the alkene.

(ii) The bond breaks, forming a carbocation inde intermediate.

(iii) A nucleophile (like Br^-) attacks the carbocation, completing the reaction.

Example:- HBr adds to an alkene, forming an alkyl halide.

Free Radical Addition:-

(i) Peroxides generate free radicals (ex. $Br^{\cdot}$).

(ii) A radical adds to the alkene, forming a carbon radical.

(iii) The carbon radical reacts with another halogen molecule, producing the final product.

Example: In the presence of peroxides, HBr adds to an alkene via a radical mechanism.

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Q5 Define elimination reaction with example?

Ans Elimination reaction involve the removal of two atoms or groups from a molecule, forming a double bond.

Types E.1 (Unimolecular elimination). A carbocation forms as the leaving as the leaving group departs, followed by deprotonation to form a double bond.

E2 (Bimolecular Elimination): The leaving group & a proton are removed simultaneously in a single step to form a double bond.

Example

E2 → $CH_3CH_2CH_2Br + KOH \rightarrow CH_2 = CH_2 + KBr + H_2O$

A hydrogen is removed from the β carbon, the bromine leaves, forming ethylene (CH_2

E_1 ex. $CH_3CH_2OH \rightarrow CH_2 = CH_2 + H_2O$ (in the presence of H_2SO_4)

- Water is lost, forming a carbocation intermediate then a proton is removed to form ethylene

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Q6 What is markonikov & anti markonikov rule, explain saytzef rule.

Ans Markovnikov's Rule:- When HX (ex. HBr) adds to an unsymmetrical alkene, the hydrogen (H) adds to the carbon with more hydrogen and the halide (X) adds to the carbon with fewer hydrogens.

example:- In the addition of HBr to propene.

$$CH_3-CH=CH_2 + HBr \rightarrow CH_3-CH(Br)-CH_3$$

Anti markovnikov's Rule:- In the presence of peroxides, the hydrogens (H) adds to the carbons with fewer hydrogens, and the halide (X) adds to the carbon with more hydrogens.

Examples with ~~peroxides~~, peroxides, the addition of HBr to propene gives

$$CH_3-CH=CH_2 + HBr \xrightarrow{Peroxide} CH_3-CH(Br)-CH_3$$

Seytzeff's (zeitsev's) Rule:- In elimination reaction, the more substituted alkene (more alkyl groups) is the major product.

example In elimination of HBr from 2-Bromobutane, the major product is ~~but~~ but-2-ene (more substituted)